A. Cut the Array

time limit per test 2 seconds
memory limit per test 512 megabytes
input standard input
output standard output

Rating: 800 | Tags: brute force, constructive algorithms, math, number theory |
https://codeforces.com/problemset/problem/2144/A

You are given an array of n non-negative integers $[a_1, a_2, \ldots, a_n]$.

Your task is to cut it into three non-empty parts: a prefix, a middle part, and a suffix. Formally, you need to choose two integers l and r such that $1 \le l$ lt r lt n, and obtain three parts:

- the prefix up the element at index l inclusive (i.e., $[a_1, a_2, \ldots, a_l]$);
- the central part from the element at index $l+1$ to the element at index r inclusive (i.e., $[a_{l+1}, a_{l+2}, \ldots, a_r]$);
- the suffix from the element at index $r+1$ to n inclusive (i.e., $[a_{r+1}, a_{r+2}, \ldots, a_n]$).

Let s_1, s_2, s_3 be the remainders of the sums of these parts modulo 3. In other words:

- $s_1 = (\sum_{i=1}^{l} a_i) \bmod 3$;
- $s_2 = (\sum_{i=l+1}^{r} a_i) \bmod 3$;
- $s_3 = (\sum_{i=r+1}^{n} a_i) \bmod 3$.

Your task is to find such boundaries l and r that either all numbers s_1, s_2, s_3 are different, or all numbers s_1, s_2, s_3 are the same.

Input

The first line contains a single integer t ($1 \le t \le 1000$) — the number of test cases.

Each test case consists of two lines:

- the first line contains a single integer n ($3 \le n \le 40$);
- the second line contains n integers $a_1, a_2, \ldots, a_n$ ($0 \le a_i \le 40$).

Output

For each test case, if a suitable pair of integers l and r ($1 \le l$ lt r lt n) exists, output these two integers (if there are multiple suitable pairs, you can output any of them). Otherwise, output two integers equal to 0.

Example

Input

```
4
6
1 2 3 4 5 6
4
1 3 3 7
3
2 1 0
5
7 2 6 2 4
```

Output

```
3 5
0 0
1 2
2 4
```

Note

Consider the examples from the statement:

- in the first example, the array is cut into parts [1, 2, 3], [4, 5], [6]; $s_1 = s_2 = s_3 = 0$;
- in the second example, there is no suitable cut;
- in the third example, the array is cut into parts [2], [1], [0]; $s_1 = 2$, $s_2 = 1$, $s_3 = 0$;
- in the fourth example, the array is cut into parts [7, 2], [6, 2], [4]; $s_1 = 0$, $s_2 = 2$, $s_3 = 1$.